```
vboxuser@Parul:~$ jobs
vboxuser@Parul:~$ sleep 100
^Z
[1]+  Stopped                 sleep 100
vboxuser@Parul:~$ jobs
[1]+  Stopped                 sleep 100
vboxuser@Parul:~$ sleep 300 &
[2] 7781
vboxuser@Parul:~$ sleep 100 &
[3] 7784
vboxuser@Parul:~$ jobs
[1]+  Stopped                 sleep 100
[2]   Running                 sleep 300 &
[3]-  Running                 sleep 100 &
vboxuser@Parul:~$ kill %2
[2]   Terminated              sleep 300
vboxuser@Parul:~$ kill %3
bash: kill: (7784) - No such process
[3]-  Done                    sleep 100
vboxuser@Parul:~$ jobs
[1]+  Stopped                 sleep 100
vboxuser@Parul:~$ sleep 5
^Z
[2]+  Stopped                 sleep 5
vboxuser@Parul:~$ jobs
[1]-  Stopped                 sleep 100
[2]+  Stopped                 sleep 5
vboxuser@Parul:~$ bg
[2]+ sleep 5 &
[2]-  Done                    sleep 5
vboxuser@Parul:~$ jobs
[1]+  Stopped                 sleep 100
vboxuser@Parul:~$ bg
[1]+ sleep 100 &
[1]+  Done                    sleep 100
vboxuser@Parul:~$ jobs
vboxuser@Parul:~$ sleep 200 &
[1] 7850
vboxuser@Parul:~$ ps
    PID TTY          TIME CMD
   7624 pts/0    00:00:00 bash
   7850 pts/0    00:00:00 sleep
   7852 pts/0    00:00:00 ps
vboxuser@Parul:~$ ps -e
```

```
vboxuser@Parul:~$ sudo nice -n -10 sleep 300 &
[5] 7304
vboxuser@Parul:~$ jobs
[1]-  Stopped                 sleep 100
[3]   Running                 sleep 500 &
[4]   Running                 nice -n 10 sleep 300 &
[5]+  Stopped                 sudo nice -n -10 sleep 300
vboxuser@Parul:~$ kill %1

[1]-  Stopped                 sleep 100
[1]-  Terminated              sleep 100
vboxuser@Parul:~$ kill %2
bash: kill: %2: no such job
vboxuser@Parul:~$ sleep 100 &
[6] 7316
vboxuser@Parul:~$ renice 10 -p 7316
7316 (process ID) old priority 0, new priority 10
vboxuser@Parul:~$ top

top - 18:42:14 up  1:58,  1 user,  load average: 0.04, 0.02, 0.05
Tasks: 259 total,   1 running, 257 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.6 us,  0.5 sy,  0.0 ni, 98.7 id,  0.1 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   2200.2 total,    198.2 free,   1318.0 used,    886.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    882.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   3229 vboxuser  20   0 6215036 393044 142540 S   9.2  17.4   1:36.67 gnome-shell
   4417 vboxuser  20   0 3488048 385996 124512 S   7.6  17.1   0:32.86 ptyxis
     15 root      20   0       0      0      0 I   0.3   0.0   0:08.25 rcu_preempt
     53 root      20   0       0      0      0 I   0.3   0.0   0:02.00 kworker/u26:0-events_unbound
   4424 vboxuser  20   0  239944   7920   7208 S   0.3   0.4   0:02.14 ptyxis-agent
   7320 vboxuser  20   0   13424   6332   4160 R   0.3   0.3   0:00.21 top
      1 root      20   0   26264  16996  11648 S   0.0   0.8   0:14.35 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.22 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u24:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.21 ksoftirqd/0
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
```

vboxuser@Parul: ~

```
vboxuser@Parul:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 16:43 ?        00:00:15 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
root           2       0  0 16:43 ?        00:00:00 [kthreadd]
root           3       2  0 16:43 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 16:43 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 16:43 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 16:43 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 16:43 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 16:43 ?        00:00:00 [kworker/R-netns]
root          12       2  0 16:43 ?        00:00:00 [kworker/u24:0-ipv6_addrconf]
root          13       2  0 16:43 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 16:43 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 16:43 ?        00:00:09 [rcu_preempt]
root          16       2  0 16:43 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 16:43 ?        00:00:01 [rcu_exp_gp_kthread_worker]
root          18       2  0 16:43 ?        00:00:00 [migration/0]
root          19       2  0 16:43 ?        00:00:00 [kprobe-optimizer]
root          20       2  0 16:43 ?        00:00:00 [idle_inject/0]
root          21       2  0 16:43 ?        00:00:00 [cpuhp/0]
root          22       2  0 16:43 ?        00:00:00 [cpuhp/1]
root          23       2  0 16:43 ?        00:00:00 [idle_inject/1]
root          24       2  0 16:43 ?        00:00:00 [migration/1]
root          25       2  0 16:43 ?        00:00:00 [ksoftirqd/1]
root          27       2  0 16:43 ?        00:00:00 [kworker/1:0H-kblockd]
root          28       2  0 16:43 ?        00:00:00 [cpuhp/2]
root          29       2  0 16:43 ?        00:00:00 [idle_inject/2]
root          30       2  0 16:43 ?        00:00:00 [migration/2]
root          31       2  0 16:43 ?        00:00:00 [ksoftirqd/2]
root          33       2  0 16:43 ?        00:00:00 [kworker/2:0H-kblockd]
root          34       2  0 16:43 ?        00:00:00 [cpuhp/3]
root          35       2  0 16:43 ?        00:00:00 [idle_inject/3]
root          36       2  0 16:43 ?        00:00:00 [migration/3]
root          37       2  0 16:43 ?        00:00:00 [ksoftirqd/3]
root          40       2  0 16:43 ?        00:00:00 [cpuhp/4]
root          41       2  0 16:43 ?        00:00:00 [idle_inject/4]
root          42       2  0 16:43 ?        00:00:00 [migration/4]
root          43       2  0 16:43 ?        00:00:00 [ksoftirqd/4]
root          46       2  0 16:43 ?        00:00:00 [cpuhp/5]
root          47       2  0 16:43 ?        00:00:00 [idle_inject/5]
root          48       2  0 16:43 ?        00:00:00 [migration/5]
root          49       2  0 16:43 ?        00:00:00 [ksoftirqd/5]
root          51       2  0 16:43 ?        00:00:00 [kworker/5:0H-kblockd]
root          53       2  0 16:43 ?        00:00:02 [kworker/u26:0-events_unbound]
root          58       2  0 16:43 ?        00:00:00 [kdevtmpfs]
```

```
vboxuser@Parul:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:15 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_gp
      5 ?        00:00:00 kworker/R-sync_wq
      6 ?        00:00:00 kworker/R-kvfree_rcu_reclaim
      7 ?        00:00:00 kworker/R-slub_flushwq
      8 ?        00:00:00 kworker/R-netns
     12 ?        00:00:00 kworker/u24:0-ipv6_addrconf
     13 ?        00:00:00 kworker/R-mm_percpu_wq
     14 ?        00:00:00 ksoftirqd/0
     15 ?        00:00:09 rcu_preempt
     16 ?        00:00:00 rcu_exp_par_gp_kthread_worker/0
     17 ?        00:00:01 rcu_exp_gp_kthread_worker
     18 ?        00:00:00 migration/0
     19 ?        00:00:00 kprobe-optimizer
     20 ?        00:00:00 idle_inject/0
     21 ?        00:00:00 cpuhp/0
     22 ?        00:00:00 cpuhp/1
     23 ?        00:00:00 idle_inject/1
     24 ?        00:00:00 migration/1
     25 ?        00:00:00 ksoftirqd/1
     27 ?        00:00:00 kworker/1:0H-kblockd
     28 ?        00:00:00 cpuhp/2
     29 ?        00:00:00 idle_inject/2
     30 ?        00:00:00 migration/2
     31 ?        00:00:00 ksoftirqd/2
     33 ?        00:00:00 kworker/2:0H-kblockd
     34 ?        00:00:00 cpuhp/3
     35 ?        00:00:00 idle_inject/3
     36 ?        00:00:00 migration/3
     37 ?        00:00:00 ksoftirqd/3
     40 ?        00:00:00 cpuhp/4
     41 ?        00:00:00 idle_inject/4
     42 ?        00:00:00 migration/4
     43 ?        00:00:00 ksoftirqd/4
     46 ?        00:00:00 cpuhp/5
```

```
vboxuser@Parul:~$ sleep 100 &
[1] 7263
vboxuser@Parul:~$ kill -15 7263
[1]+  Terminated              sleep 100
vboxuser@Parul:~$ sleep 100 &
[1] 7274
vboxuser@Parul:~$ kill -9 7274
[1]+  Killed                  sleep 100
vboxuser@Parul:~$ sleep 100 &
[1] 7279
vboxuser@Parul:~$ kill -STOP 7279

[1]+  Stopped                 sleep 100
vboxuser@Parul:~$ sleep 100 &
[2] 7282
vboxuser@Parul:~$ kill -CONT 7282
vboxuser@Parul:~$ sleep 500 &
[3] 7286
[2]   Done                    sleep 100
vboxuser@Parul:~$ ps -o pid,ni,cmd
    PID  NI CMD
   7203   0 /usr/bin/bash
   7279   0 sleep 100
   7286   0 sleep 500
   7288   0 ps -o pid,ni,cmd
vboxuser@Parul:~$ nice -n 10 sleep 300 &
[4] 7290
vboxuser@Parul:~$ ps -o pid,ni,cmd
    PID  NI CMD
   7203   0 /usr/bin/bash
   7279   0 sleep 100
   7286   0 sleep 500
   7290  10 sleep 300
   7293   0 ps -o pid,ni,cmd
```

```
vboxuser    7850    7624  0 19:20 pts/0    00:00:00 sleep 200
vboxuser    7861    7624 99 19:22 pts/0    00:00:00 ps -ef
vboxuser@Parul:~$ fg
sleep 200
^Z
[1]+  Stopped                 sleep 200
vboxuser@Parul:~$ sleep 200 &
[2] 7874
vboxuser@Parul:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.7  26264 16996 ?        Ss   16:43   0:15 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
root           2  0.0  0.0      0     0 ?        S    16:43   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    16:43   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-netns]
root          12  0.0  0.0      0     0 ?        I    16:43   0:00 [kworker/u24:0-ipv6_addrconf]
root          13  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    16:43   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    16:43   0:09 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    16:43   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    16:43   0:01 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    16:43   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    16:43   0:00 [kprobe-optimizer]
root          20  0.0  0.0      0     0 ?        S    16:43   0:00 [idle_inject/0]
root          21  0.0  0.0      0     0 ?        S    16:43   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    16:43   0:00 [cpuhp/1]
root          23  0.0  0.0      0     0 ?        S    16:43   0:00 [idle_inject/1]
root          24  0.0  0.0      0     0 ?        S    16:43   0:00 [migration/1]
root          25  0.0  0.0      0     0 ?        S    16:43   0:00 [ksoftirqd/1]
root          27  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/1:0H-kblockd]
root          28  0.0  0.0      0     0 ?        S    16:43   0:00 [cpuhp/2]
root          29  0.0  0.0      0     0 ?        S    16:43   0:00 [idle_inject/2]
root          30  0.0  0.0      0     0 ?        S    16:43   0:00 [migration/2]
root          31  0.0  0.0      0     0 ?        S    16:43   0:00 [ksoftirqd/2]
root          33  0.0  0.0      0     0 ?        I<   16:43   0:00 [kworker/2:0H-kblockd]
root          34  0.0  0.0      0     0 ?        S    16:43   0:00 [cpuhp/3]
root          35  0.0  0.0      0     0 ?        S    16:43   0:00 [idle_inject/3]
root          36  0.0  0.0      0     0 ?        S    16:43   0:00 [migration/3]
root          37  0.0  0.0      0     0 ?        S    16:43   0:00 [ksoftirqd/3]
root          40  0.0  0.0      0     0 ?        S    16:43   0:00 [cpuhp/4]
root          41  0.0  0.0      0     0 ?        S    16:43   0:00 [idle_inject/4]
```